SST TAWG

Technical Advisory Working Group

Meeting No. 21

Ted Bohn, Mack Knobbe,
Alex McDonald, Rakib Rahman

28 MAY 2026

Introduction

- Alex McDonald, Mack Knobbe, Ted Bohn – Organizers, Facilitators

- This is a public gathering of individuals and groups interested in the subject matter of power electronics and AC/DC conversion devices focused on DC power delivery for suitable purposes, such as large-scale AI/Data centers (and EV charging plazas…).

- This is not a standards body, and it has no regulatory authority

- Persons attending and communicating here are volunteering of free will and in the interest of mutual free public collaboration

- **Meeting Recording Policy in 2026**: Each presenter/speaker has right to decline recording/archive of meeting presentations. Recording for today is at https://anl.box.com/s/k4g27xkiky2a40pzwuwt7hyead0kxmwv

Agenda

- Introduction and Scope Discussion Ted 5 min
- Previous meeting recap; (Critical Loop, standards, SST events) Ted 5 min
- Current Events; Conferences, power modules, white papers, articles, standards, Enphase, Delta, Eaton examples Ted 25 min
- Main Topic: ~30 min
  - **Ashish Kumar- Wolfspeed**; 10kV SiC devices and modules for SSTs
  - **Ram Adapa- EPRI**; *AC line conversion to DC or state of the art in HVDC technology*
- Group Discussion
- Next Meeting June 25th (Schneider, Proterial magnetics?)

Gaps that we have been addressing:

- Technical Specifications
- Standards – who, which?
- Certification – safety, comms, cyber, EMT-conducted emissions issues
- Testing
  - Safety testing (Specialists?)
  - Functional testing (National Labs?)
  - Application specific testing (utility)
- Demonstrations, Pilots, Collaborations
- What does it take to get to market maturity for commercial off the shelf certified and interoperable systems?

Recap of April SST Meeting: Ted Bohn; Industry snapshot

- ***Summary of SST related meetings, standards references, Critical Loop presentation-combiners***
- – Presentations https://anl.box.com/s/nhn37blt00ztu7l1qwhfqcw2wltzp0mr

  - Meeting Agenda Slides https://anl.box.com/s/6doasrhl0hd59s5jc87se9hhbnohchli

  - SST (ANL led) White Paper Drafts https://anl.box.com/s/xdx6ho0rvkop6ltqljrbfqlj55ghoxow

# of ~500	Category
11	Connector
14	Consultant
8	DC Dsitribution
57	EVSE
25	EVSP
35	Lab-Agency
3	Load ctrl, Mining
23	OEM
6	Operator
20	Organizations
44	Planning Services
10	Power Elec.
3	Rail
5	Sales
29	SST Manufacturer
4	Systems Engineers
9	Testing
21	University
62	Utility

Topics of Subteams:

- **Standards:**
- **Safety:** OSHA, IEC, other?
- **Data Collection:** Confidentiality
- **Testing:** Safety, Functional
- **Certification:** AHJs
- **White Paper/Specifications Template**

List of SST Manufacturers and/or Demonstration Projects

ABB, Hitachi, Siemens, Vernova, Eaton, General Electric, Schneider Electric, Mitsubishi, and Delta Electronics.
*Many **are actively partnered with Nvidia** power system component providers.*

Item	Name	Country/state	URL
1	AK Power Solutions	US/VA, India	https://akpowers.com/24weeks/
2	Alderbuck	US/CA	https://alderbuck.com/#
3	Amperesand	Thailand/CA	https://amperesand.io/
4	Charge America(WattEV)	US/CA	https://www.chargeamerica.us/
5	DC Grid	US/CA	https://dcgrid.io/
6	Delta Electronics	Taiwan/CA	https://www.deltaww.com/en-US/products/solid-state-transformer/ALL/
7	DG Matrix	US/NC	https://dgmatrix.com/power-router-technology
8	Eaton	US/WI/NC	https://www.eaton.com/us/en-us/company/news-insights/news-releases/2021/eaton-receives-u-s--department-of-energy-award.html
9	Enphase	US/CA	https://enphase.com/iq-solid-state-transformer
10	G.E. Vernova	US/NY	https://www.gevernova.com/power-conversion/product-solutions/microgrid-solutions-for-data-center
11	Gridbridge ERMCO	US/NC	https://grid-bridge.com/products/
12	Heron Power	US/CA	https://www.heronpower.com/
13	Hyperscale Power	Switzerland	https://hyperscale-power.com/
14	Infinity Flow	US/TX	https://infmiles.com/breakthrough-achievement-sst-module/
15	Midwest E- Propulsion	US/WI	https://www.linkedin.com/in/ajith-wijenayake-54735819/
16	Mission Power	US/NY	https://www.missionpower.com/products
17	Novos Power	US/San Diego	https://www.novospower.com/
18	Polyax Inc.	US/TX	https://polyax.com/#top
19	Resilient Power	US/TX	https://www.resilientpower.com/commercial-applications
20	Rocketruck	US/CA	https://rocketruck.com/solidstatetransformers/
21	Solid State Power	US/TX	https://solidstatepower.net/page5.html#header2-a
(Univ)	NC State	US/NC	https://freedm.wordpress.ncsu.edu/sst

SST TAWG Welcomes Enphase to the Group

- Enphase has a good 20 page whitepaper on SSTs/their product
  https://enphase.com/download/iq-sst-white-paper

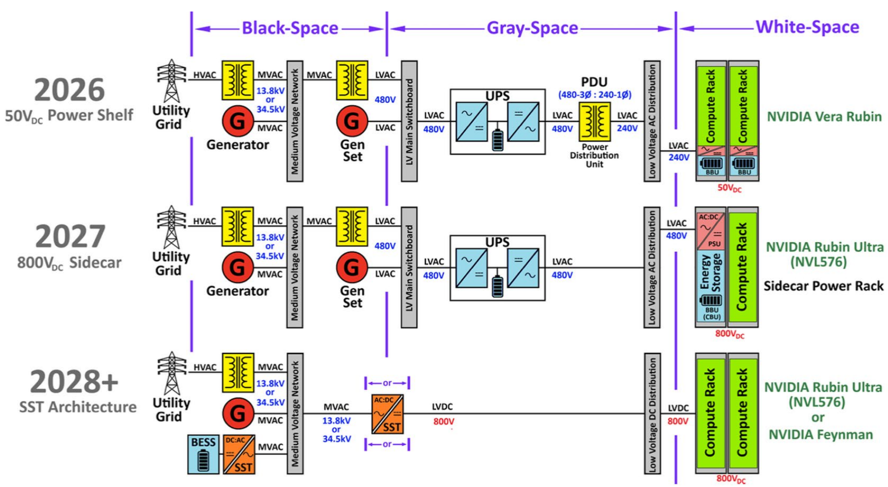


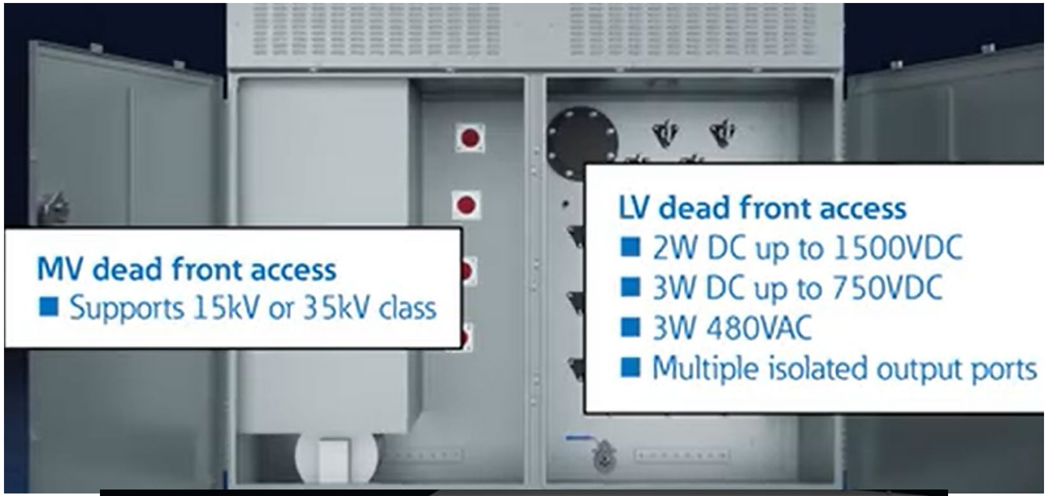
Figure 2: Line diagrams depicting typical AI data center system configurations: 1) Today, 2) Mid-2027, and 3) 2028+

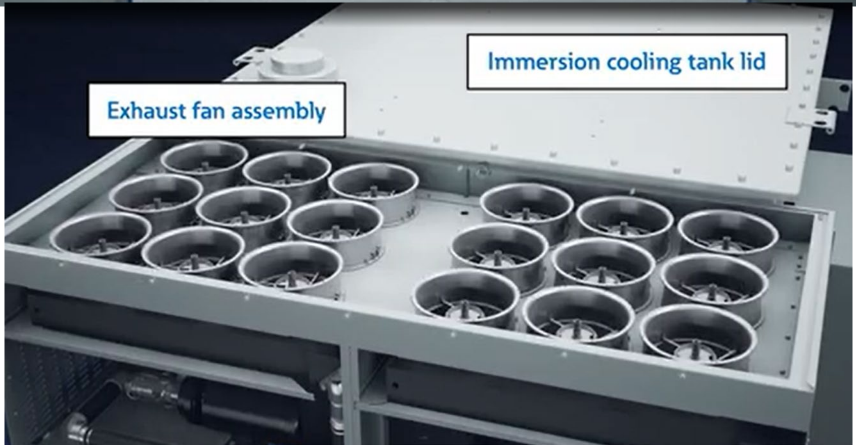
Intelligent power for AI.

1.25 MW	99.999%	<1 ms	342
PER RACK · SCALES TO 5 MW	SYSTEM AVAILABILITY	TRANSIENT RESPONSE	POWER MODULES PER RACK

SST TAWG Brochures Folder; Eaton Example

- Brochure folder added to SST Box folders: https://anl.box.com/s/6h7elf0q5sp5nlhtgmf6y6gsuowdxznr
- Eaton Medium Voltage Solid State Transformer: 2MW, 15kV class, 97% eff., 12 Isolated output connections; 98A AC input, .97 PF, <5% THD, IP69, 15k lbs
  https://www.eaton.com/us/en-us/catalog/medium-voltage-power-distribution-control-systems/medium-voltage-solid-state-transformer.html



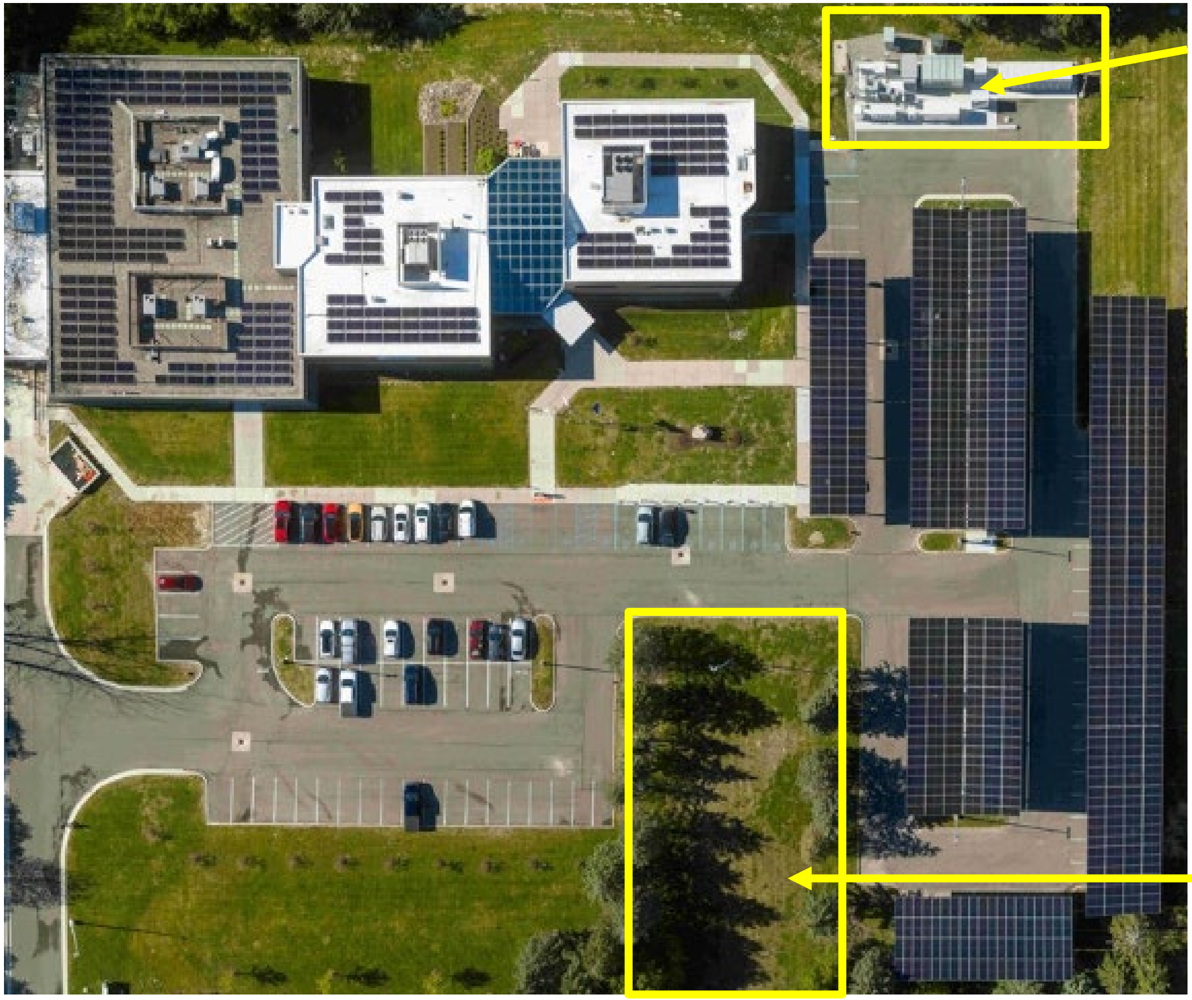


Next Steps

- Next meeting June 25th, 2026 (Schneider, Proterial?)
- Open guest speaker topic **for July meeting**?
  - Other speakers interested?

Delta Electronics North America SST Testbed

- https://www.solarpowerworldonline.com/2026/05/delta-electronics-on-site-microgrid-to-provide-valuable-solar-storage-insights/
- ***Delta Electronics' on-site microgrid to provide valuable solar + storage insights***



Microgrid to build on this foundation with five energy storage circuits:
- two back-to-back 5-MWh units to be installed by year's end and configured to emulate large electrical loads, allowing Delta to test
- black-start scenarios
- off-grid switching
- grid-loss recovery without requiring a physical megawatt-scale load on site.
- 300-kW gas turbine is planned to add the ability to study how solar, storage and gas generation interact as grid conditions change.

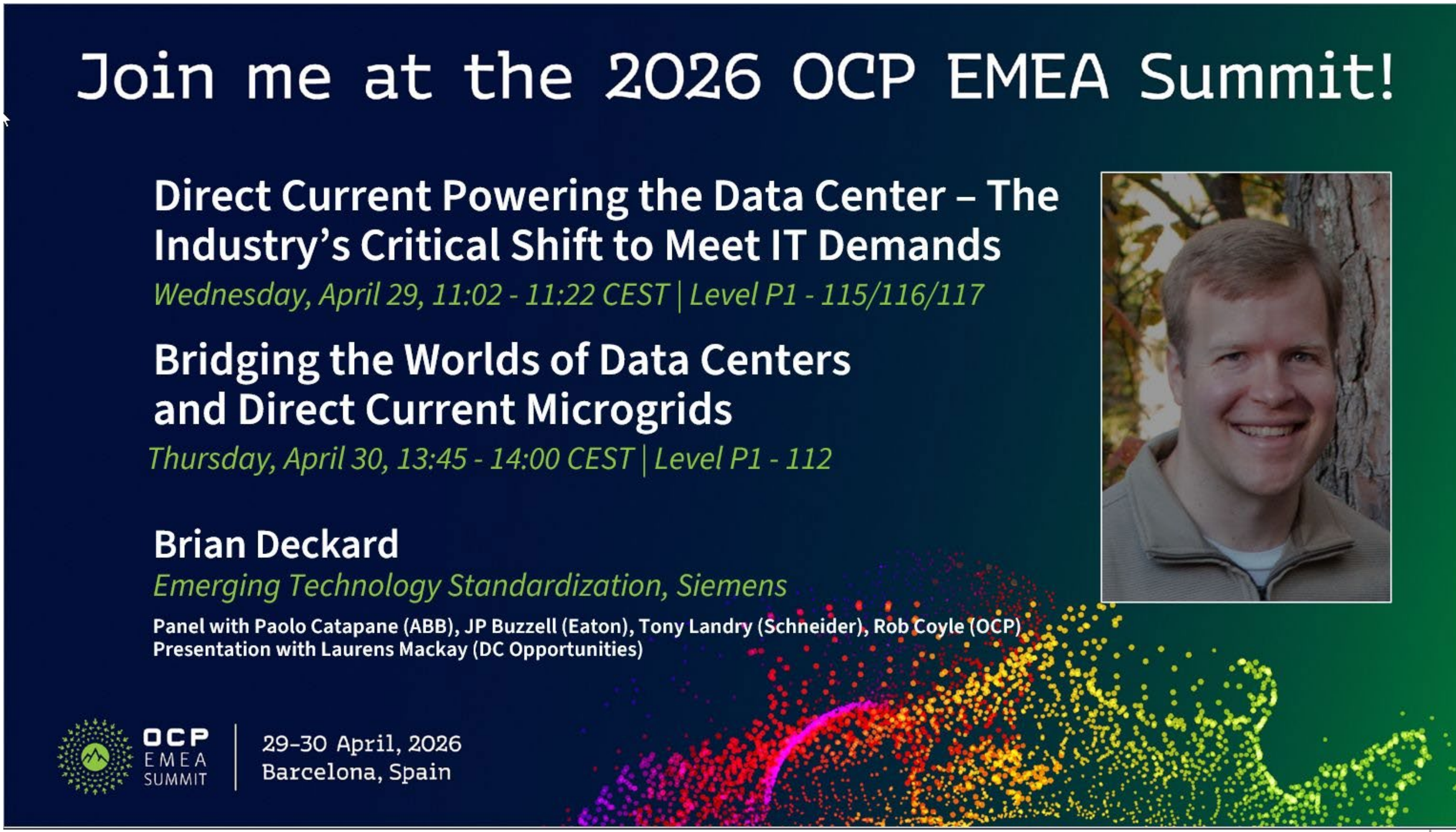
Delta Electronics North America SST Testbed

- Delta's facility onsite microgrid integrates
  - 425 kW of solar generation
  - eight Level 2 EV chargers
  - 400-kW DC fast EV charger 2/Delta Gen 1 SST
  - 13.2-kV medium-voltage interconnection/transformer
  - Delta's 3-MW power conditioning system (PCS) and 2.8 MWh of stationary BESS
  -Delta's energy management system (EMS) and in-house VTScada SCADA platform

Ph

Open Compute Project EMEA Summit, April 29-30

- https://www.opencompute.org/summit/emea-summit



Open Compute Project Foundation EMEA Global Summit 2026

- April 29-30, Barcelona (Dusan Brhlik, Direct Energy Partners, LinkedIn post)
- *Next Summit is Oct 12-15, 2026 San Jose CA* *https://www.opencompute.org/summit/global-summit*

Current OS 11th Plenary Meeting- RWTH Aachen Germany

- May 26-28th, https://currentos.org/
- **Data Center Focus:** The plenary features the LVDC Data Center Summit, focusing on addressing emerging needs in IEC standardization for high-voltage DC architectures.
- **General Assembly**: The Plenary and General Assembly provide a collaborative forum for industry and research leaders from organizations like Schneider Electric, Nexans, and FEN Research Campus to share real-world deployment data and reference architectures.
- **Global Community:** Current/OS connects over 130 companies globally working to make electrical grids more efficient by reducing AC/DC conversion losses.

Solid-State Battery Summit Plus Co-Located Battery Safety Summit

- *Aug 11-12, 2026, Chicago; https://battery-tech.net/battery-event/solid-state-battery-summit-2026/*
- Offering meaningful increases in energy density and significant improvements in safety over conventional liquid-electrolyte lithium-ion systems, these next-generation battery architectures, particularly semi-solid-state designs are emerging as a scalable bridge toward fully solid-state solutions.
- 6th year, the **Solid-State Battery Summit** will present the global solid-state and semi-solid-state battery ecosystem from multiple perspectives, including advances in materials chemistry, cell and pack engineering, manufacturing scalability, and safety performance, as well as cost-reduction strategies being pursued by leading developers and OEMs.
- ...forward-looking market outlook covering forecasted adoption and commercialization timelines across China, Japan, Korea, Europe, and the United States.
- Co-located with 16th Annual **Battery Safety Summit**, don't miss your opportunity to gain critical insights into the latest technical, manufacturing, and commercial developments from the major global players advancing safe solid-state and semi-solid-state battery technologies.

YOTTA Powering AI Conference, Sept 28-30 Las Vegas

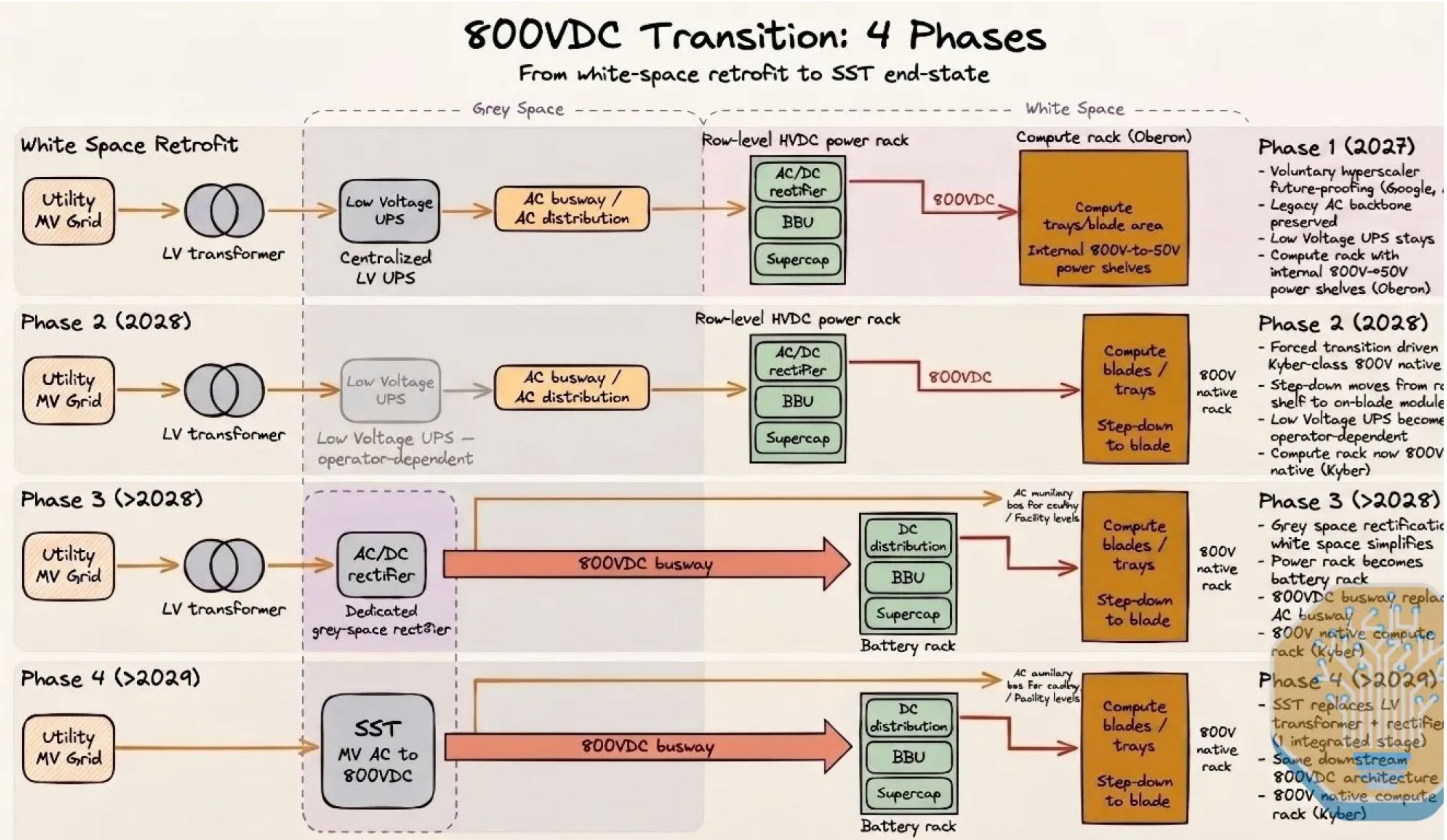
- https://www.yotta-event.com/2026-speakers/
- 6000 attendees, 350 speakers,
- *Yotta 2026 is where the future of energy for AI infrastructure comes into focus. We're on track to welcome 350+ speakers this September, with new names being added every day from across utilities, clean tech, infrastructure and capital. From behind-the-meter strategies and microgrids to grid evolution and new energy investment models, Yotta brings together the conversations redefining how power is generated, delivered and scaled for AI.*
- *Leaders from companies like Schneider Electric, ON.energy, Giga Energy, Xcel Energy, Exowatt, Engie, DG Matrix, EmeraldAI, Sightlight Climate and Idaho National Lab are already confirmed, alongside the operators, innovators and investors tackling the single biggest constraint facing AI growth: power.*

Semi Analysis Article on 800vdc/SSTs, with Industry Input

- (Multi-part series) ***Welcome to the Power Chain Roller Coaster***
  https://newsletter.semianalysis.com/p/inside-the-800vdc-revolution-part (**70 pages**, many tables/figures)
- *We'd like to thank DG Matrix, Novos Power, and Aran Industries for their contributions and insights during the preparation of this deep dive.*

Semi Analysis Article on 800vdc/SSTs, with Industry Input

- 70 pages of figures, dates, tables, etc



Ashish Kumar- Wolfspeed

10kV SiC Devices and Modules for SSTs

Design nuances	MV-rated SiC power devices <5kV blocking voltage	HV-rated SiC power devices >5kV blocking voltage
Architecture	Series stacking of multiple devices, often in three to five-level architectures	Wolfspeed 10kV die can enable a simple two-level cell
Modularity	Smaller building blocks are easier to handle and service	Larger and less modular therefore not as easy to handle
Insulation	Requires precise attention to insulation coordination between cells	Simpler insulation
Control Scheme	Control algorithm needed to gate the series switches for a cleaner output	Less complex control scheme due to fewer cells
Gate Driver	3.3kV rated SiC gate drivers are commercially available	Very few compatible solutions available above 3.3 kV, requiring vendors to do custom development and qualification
Electromagnetic Interference (EMI)	Multi-level architecture requires smaller dv/dt	Switch with high dv/dt requiring ultra-low-parasitic design

Table 1: Key nuances between SSTs designed using MV and HV-rated SiC devices

Ram Adapa- EPRI

AC line conversion to DC or state of the art in HVDC technology